

Troubleshooting Guide

Common Setup Issues & Solutions

Error: Cannot find module '@tailwindcss/typography'

Problem: Missing Tailwind CSS typography plugin

Solution:

```
cd frontend
npm install @tailwindcss/typography
```

Or run the quick fix script:

```
chmod +x quick-fix.sh
./quick-fix.sh
```

Error: cargo: not found or Rust not installed

Problem: Rust toolchain not installed

Solution:

```
# Install Rust
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh

# Restart terminal, then verify
cargo --version
```

Error: TURSO_DATABASE_URL must be set

Problem: Missing environment variables

Solution:

```
# 1. Create Turso database
turso db create blog-db

# 2. Get credentials
turso db show blog-db --url
turso db tokens create blog-db

# 3. Create backend/.env
cp backend/.env.example backend/.env
```

```
# 4. Edit backend/.env with your credentials
TURSO_DATABASE_URL=libsql://your-db.turso.io
TURSO_AUTH_TOKEN=your-token-here
```

Error: Compilation errors in Rust

Problem: Missing dependencies or version mismatch

Solution:

```
cd backend
```

```
# Clean build
```

```
cargo clean
```

```
# Update dependencies
```

```
cargo update
```

```
# Rebuild
```

```
cargo build
```

Error: Module not found in Next.js

Problem: Node modules not installed

Solution:

```
cd frontend
```

```
# Remove existing modules
```

```
rm -rf node_modules package-lock.json
```

```
# Clean install
```

```
npm install
```

```
# Try again
```

```
npm run dev
```

Error: turso: command not found

Problem: Turso CLI not installed

Solution:

```
# macOS/Linux
curl -sSfL https://get.tur.so/install.sh | bash

# Restart terminal, then verify
turso --version
```

Error: Port already in use (Backend)

Problem: Port 3000 already occupied

Solution 1 - Change port:

```
# In backend/.env
PORT=3001
```

Solution 2 - Kill existing process:

```
# Find process on port 3000
lsof -ti:3000
```

```
# Kill it
kill -9 $(lsof -ti:3000)
```

Error: Port already in use (Frontend)

Problem: Port 3001 already occupied

Solution:

```
# Next.js will automatically try port 3002, 3003, etc.
# Or specify a port:
PORT=3002 npm run dev
```

Error: CORS policy blocked

Problem: Frontend can't access backend

Solution:

```
# 1. Check frontend/.env.local
NEXT_PUBLIC_API_URL=http://localhost:3000
```

```
# 2. Verify backend CORS is configured (already done)
# 3. Make sure both are running
```

Error: Images won't upload

Problem: Various image upload issues

Solution 1 - Check file size:

Max size is 5MB per image
Compress large images before upload

Solution 2 - Check file type:

Allowed: JPG, PNG, GIF, WebP
Not allowed: SVG, BMP, TIFF

Solution 3 - Check database:

```
# Verify database connection
turso db list
turso db show blog-db
```

Error: Database schema errors

Problem: Schema not initialized

Solution:

```
cd backend

# Run setup script
cargo run --bin setup

# This will:
# - Initialize schema
# - Create admin user
# - Set up tables
```

Error: Authentication not working

Problem: No admin user or wrong credentials

Solution:

```
# Create admin user
cd backend
cargo run --bin setup

# Default credentials:
# Email: admin@example.com
```

Password: password123

Change password after first login!

Error: TypeScript errors in frontend

Problem: Type mismatches

Solution:

`cd frontend`

Check TypeScript

`npm run build`

If issues persist, check:

- All imports are correct

- Component props match interfaces

- API response types match

Error: Failed to connect to database

Problem: Invalid Turso credentials or network

Solution 1 - Verify credentials:

Test database connection

`turso db shell blog-db`

If this works, credentials are correct

Update backend/.env with correct values

Solution 2 - Check network:

Verify internet connection

Turso requires internet access

`ping turso.io`

Error: Build fails on Vercel

Problem: Missing environment variables or build errors

Solution:

```
# 1. Add environment variables in Vercel dashboard:
NEXT_PUBLIC_API_URL=https://your-backend.vercel.app
TURSO_DATABASE_URL=libsql://...
TURSO_AUTH_TOKEN=...

# 2. Verify vercel.json is present
# 3. Check build logs for specific errors
```

Debugging Tips

Check Backend Logs

```
cd backend
RUST_LOG=debug cargo run
```

Check Frontend Logs

```
cd frontend
npm run dev
# Check browser console for errors
```

Check Database

```
# Open Turso shell
turso db shell blog-db

# Check tables
.tables

# Check data
SELECT * FROM images LIMIT 5;
SELECT * FROM posts LIMIT 5;
```

Test API Endpoints

```
# Health check
curl http://localhost:3000/health

# Get posts
curl http://localhost:3000/api/posts

# Get analytics
curl http://localhost:3000/api/images/analytics
```

Still Having Issues?

1. Run Quick Fix Script

```
chmod +x quick-fix.sh
./quick-fix.sh
```

2. Check Documentation

- README.md - Overview
- SETUP_INSTRUCTIONS.md - Detailed setup
- QUICKSTART.md - 5-minute guide

3. Verify Installation

```
# Check versions
node --version    # Should be 18+
npm --version     # Should be 9+
cargo --version   # Should be 1.70+
turso --version   # Should be latest
```

4. Clean Install

```
# Backend
cd backend
cargo clean
cargo build

# Frontend
cd frontend
rm -rf node_modules package-lock.json
npm install
```

Common Setup Mistakes

Running in wrong directory

```
# Wrong:
npm run dev # From root
```

```
# Correct:
cd frontend
npm run dev
```

Missing .env files

```
# Need both:
backend/.env          # Backend config
frontend/.env.local  # Frontend config
```

Not running both servers

```
# Need BOTH running:
# Terminal 1:
cd backend && cargo run

# Terminal 2:
cd frontend && npm run dev
```

Using wrong ports

```
# Backend: http://localhost:3000
# Frontend: http://localhost:3001
# Make sure NEXT_PUBLIC_API_URL points to 3000
```

Verification Checklist

After setup, verify everything works:

- ☐ Backend starts without errors
 - ☐ Frontend starts without errors
 - ☐ Can login at /login
 - ☐ Can access /admin
 - ☐ Can create a post
 - ☐ Can upload an image
 - ☐ Can view analytics
 - ☐ Can search images
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Quick Start Commands

```
# Initial Setup
./quick-fix.sh

# Start Backend
cd backend
cargo run

# Start Frontend (new terminal)
```



```
cd frontend  
npm run dev
```

```
# Create Admin User  
cd backend  
cargo run --bin setup
```

Need more help? Check the detailed guides: - `SETUP_INSTRUCTIONS.md` -
`BUG_FIXES_AND_ROADMAP.md` - Individual feature guides (`IMAGE_OPTIMIZATION_GUIDE.md`,
etc.)